

Materials Bridge Line A: Ferron Coherence Recoverability Probe for HAOS-IIP

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Abstract

This numbered update records release 64.1: Materials Bridge Line A for HAOS-IIP. The bridge is an external-data sidecar experiment over published ferron / NbOI₂ source data, not a frozen-core modification and not a proof of HAOS-IIP. The runner downloads the Figshare source-data record for Choe et al., “Observation of coherent ferron emission and propagation,” parses three public XLSX files, extracts time traces and spectra, and applies bounded HAOS-style recoverability proxies to coherent polarization-wave readouts. The validation status is **PASS** as an external-data bridge: **data_status=REAL_DATA_LOADED**, three raw source files are hashed and preserved, nine time traces and six spectra or peak records are parsed, and the recoverability sweep runs deterministically. A second-pass spectral audit finds the target-window feature in 15/15 audited spectral records near 3.13 THz, with mean peak frequency 3.13893 THz, mean spectral recoverability 0.928455, and no detected k_* collapse under the current proxy thresholds. A raw STFT/time-frequency audit inspects the XLSX files and returns **NO_STFT_DATA_FOUND**: one STFT-like sheet is present, but it lacks a usable frequency axis or validated time-frequency grid, so no raw STFT recoverability claim is made. The same Fig. 2d sheet is parsed separately as published post-processed target-band STFT intensity traces: eight traces and 216 time points support a bounded propagation audit with group-velocity proxies of 58.8 km/s for 96 nm and 100 km/s for 224 nm. A derived, computed-from-time-trace STFT diagnostic over the raw Fig. 2a/2b traces gives mean envelope correlation 0.995318 against the published target-band traces and mean computed group velocity 101754 m/s. A final unified persistence ledger reports **ferron_persistence_proxy=0.957192** while preserving **raw_stft_time_frequency_grid_status=NO_STFT_DATA_FOUND**. The safe claim is that HAOS-IIP now has one external materials-data bridge pipeline that touches real published material-coherence data, records provenance, runs bounded telemetry, and can say both yes and no without overclaiming.

Contents

1	Status and Scope	2
2	Strict Non-Claims	3
3	Public Data Source	3
3.1	Downloaded Raw Files	3
4	HAOS Translation	4
5	Metric Definitions	4

6	Validation Result	5
7	Distance-Series Read	5
8	Second-Pass Spectral Feature Audit	6
9	Raw STFT / Time-Frequency Audit	7
10	Published STFT Target-Band Intensity Trace Audit	7
11	Computed From Time-Trace STFT Diagnostic	8
12	Unified Harmonic-Address Persistence Summary	9
13	Figures	10
14	Interpretation	10
15	Artifacts	11
16	Conclusion	12

1 Status and Scope

This paper is release 64.1 in the HAOS-IIP numbered paper sequence. It records:

- a new external-data sidecar experiment;
- public Figshare data ingestion;
- raw-file preservation and SHA256 manifesting;
- conservative parsing of source-data XLSX workbooks;
- bounded recoverability telemetry over ferron / NbOI₂ coherent readouts;
- a second-pass spectral feature audit near 3.13 THz;
- a raw STFT/time-frequency audit that records absence of usable STFT grids without fabricating data;
- a bounded published target-band STFT intensity trace audit over Fig. 2d;
- a derived computed-from-time-trace STFT diagnostic over raw Fig. 2a/2b traces;
- a unified ferron harmonic-address persistence summary that preserves the raw-grid boundary;
- a validation result that preserves the absence of collapse.

The sidecar path is:

```
experiments/materials_bridge/ferron_coherence_demo/
```

The bridge does not modify:

- `haos_core`;
- frozen phase folders;
- frozen telemetry definitions;
- theory files;
- existing experiments;
- prior paper releases.

2 Strict Non-Claims

This release does not claim:

- that ferrons prove HAOS-IIP;
- that ferrons embody HAOS-IIP;
- a new materials theory;
- a quantum-consciousness statement;
- an Orch-OR statement;
- that recoverability proxies are final physical observables;
- that a missing k_* is a universal coherence guarantee.

The narrower question is:

Can HAOS-style recoverability metrics be applied to published ferron / NbOI₂ coherent polarization-wave data without fabricating data or forcing a positive collapse result?

3 Public Data Source

The primary external source is:

- Choe et al., “Observation of coherent ferron emission and propagation,” *Nature Materials*, 2026, DOI 10.1038/s41563-026-02597-4.
- Source data DOI: 10.6084/m9.figshare.31895293.

The secondary interpretive reference is:

- Wang et al., “Ultrafast dynamics of ferroelectric polarization of NbOI₂ captured with femtosecond electron diffraction,” *Nature Communications*, 2025, DOI 10.1038/s41467-025-63533-9.

3.1 Downloaded Raw Files

The runner downloaded and preserved three raw Figshare XLSX files:

File	Size	SHA256
SourcedataFig.1.xlsx bed3ca163043e2c386c546980b9e82f3	129205	05a38cc576816479f851c0f4dfbe6370
SourcedataFig.2.xlsx 669139e3ecc7ba73833b53227a81dc25	420687	8cba21ae3cba830a63a71a997532d892
SourcedataFig.4.xlsx 33599e5289fb25430bbad2c74f7fa93e	5194876	42ea381f87d9046fbacce64246324d7b

Raw files live under:

`experiments/materials_bridge/ferron_coherence_demo/outputs/raw/`

They are not modified by the parser.

4 HAOS Translation

The bridge maps the materials readout into HAOS-style telemetry as follows:

HAOS-facing term	Materials-side read
carrier	ferroelectric polarization order
perturbation	laser pulse, distance, thickness, fluence, temperature, boundary or disorder effects when present
recoverable structure	coherent ferron / polarization-wave mode
observable readout	narrow-band THz emission, 3.13 THz amplitude, transient reflectance oscillation, Fourier or STFT peak
visible failure	loss of narrow-band signal, spectral broadening, amplitude collapse, incoherent background dominance, or propagation failure

This is a measurement dictionary, not a materials ontology.

5 Metric Definitions

The sidecar computes bounded proxies:

- **ferron_peak_recoverability**: normalized peak amplitude times peak-frequency stability times linewidth stability;
- **propagation_recoverability**: survival of coherent amplitude across distance when distance metadata exists;
- **temporal_recoverability**: time-trace coherence proxy when delay/time traces are parsed;
- **combined_recoverability**: mean of available recoverability channels;

- ΔP : change in combined recoverability between successive conditions;
- k_* : first sustained recoverability collapse below threshold 0.70 for two consecutive steps;
- `safety_margin`: remaining distance from a condition to k_* , when k_* exists;
- `confidence`: signal clarity proxy from peak-to-background or strongest persistence drop;
- `visible_failure`: stricter readout failure flag.

The important ordering rule is:

k_* may occur before visible failure.

That is the bridge test. In this release, k_* does not occur.

6 Validation Result

Field	Value
Validation status	PASS
Data status	REAL_DATA_LOADED
External source data	10.6084/m9.figshare.31895293
Files loaded	3
Time traces parsed	9
Spectra or peak records parsed	6
STFT maps parsed	0
Baseline recoverability	1.0
k_* level	None
First visible failure level	None
Early detection	False
Spectral audit status	PARTIAL
STFT audit status	NO_STFT_DATA_FOUND
Published target-band trace status	PASS
Computed STFT diagnostic status	PASS
Unified persistence summary status	PASS

The confidence summary is:

Statistic	Value
Minimum	0.8654037886
Mean	0.9798696276
Maximum	0.9999926836

7 Distance-Series Read

The parsed Fig. 2 distance traces provide the closest match to the intended propagation probe. Two distance-series groups are parsed at 0, 2, 4, 6 μm .

Series	Distance	Peak frequency	Peak recoverability	Combined recoverability
Fig.2a	0 μm	3.132363132	1.0000000000	1.0000000000
Fig.2a	2 μm	3.132363132	1.0000000000	1.0000000000
Fig.2a	4 μm	3.139293139	0.9778761062	0.9926253687
Fig.2a	6 μm	3.139293139	0.7519864998	0.9173288333
Fig.2b	0 μm	3.132363132	1.0000000000	1.0000000000
Fig.2b	2 μm	3.132363132	1.0000000000	1.0000000000
Fig.2b	4 μm	3.139293139	0.7519864998	0.9173288333
Fig.2b	6 μm	3.139293139	0.7519864998	0.9173288333

The current thresholds do not mark a sustained collapse. This is the result: the bridge is operational, the external data is real, and the measured proxy does not force an early-failure story.

8 Second-Pass Spectral Feature Audit

The second-pass spectral audit asks a narrower question:

What spectral features stand out in the real ferron / NbOI_2 data, and do they remain recoverable under the available conditions?

The audit does not change the first-pass recoverability result. It inspects parsed spectra directly and also records spectra derived from parsed time traces as derived diagnostics, not raw STFT maps.

Field	Value
Spectral audit status	PARTIAL
Audited spectral records	15
Target-window peaks found	15/15
Target frequency	3.13 THz
Search window	± 0.20 THz
Mean peak frequency	3.13893 THz
Peak-frequency range	3.13236--3.192 THz
Mean spectral recoverability	0.928455
Linewidth proxy availability	14/15 records
Mean linewidth proxy	0.0253178 THz
Peak-to-background availability	14/15 records
Mean peak-to-background	8194.98
Spectral $k_\star$ detected	False
Visible failure detected	False

The **PARTIAL** status is intentional. It means the spectral peak audit succeeds, but one record does not support the linewidth or peak-to-background proxy. The audit does not fill those missing quantities.

The bounded read is:

A stable target-window peak near 3.13 THz appears across the audited records. Peak frequency remains stable under the available conditions. No sustained $k_\star$ collapse is detected under the current proxy thresholds.

9 Raw STFT / Time-Frequency Audit

The STFT audit asks whether the downloaded XLSX files contain a usable raw time-frequency map. The rule is strict: time traces may not be converted into synthetic STFT maps and then reported as raw STFT data.

Field	Value
STFT audit status	NO_STFT_DATA_FOUND
Raw XLSX files inspected	3
Candidate STFT-like sheets inspected	1
Accepted raw STFT maps	0
STFT time points audited	0
STFT target peak records found	0
Mean STFT recoverability	None
STFT k_* detected	False

The one STFT-like candidate is:

`SourcedataFig.2.xlsx#Fig.2d`

It is rejected because it contains time plus STFT Int columns but no usable frequency axis and no validated time-frequency amplitude grid:

`long_form_frequency_column_missing; matrix_axes_not_numeric_enough`

Therefore, this release makes no STFT recoverability claim. The absence of a parsed raw STFT grid is not treated as negative evidence against the material system; it is only a data-format boundary for this sidecar.

10 Published STFT Target-Band Intensity Trace Audit

The rejected raw-STFT candidate is still scientifically useful in a narrower form. The sheet `SourcedataFig.2.xlsx#Fig.2d` exports published, post-processed STFT Int traces in the target band. These traces are not raw time-frequency maps because no frequency axis is present, so the sidecar parses them separately as:

`published_stft_target_band_intensity_trace`

This preserves the raw-grid boundary while allowing a bounded propagation-readout audit over the published target-band envelope.

Field	Value
Published trace audit status	PASS
Mode	published_stft_target_band_intensity_trace
Source sheet	Source data Fig.2.xlsx#Fig.2d
Traces found	8
Time points audited	216
Peak records	8
Thickness groups	2
Mean group velocity	79411.8 m/s
Mean velocity consistency with 10^5 m/s	0.794118
Mean envelope recoverability	0.625
Raw STFT grid status	NO_STFT_DATA_FOUND

Thickness	Slope	Group velocity	R^2	Monotonic delay
96 nm	17 ps/um	58823.5 m/s	0.955372	1.0
224 nm	10 ps/um	100000 m/s	0.987654	1.0

The bounded read is that peak arrival time increases with distance in the parsed thickness groups. This supports a propagation-readout audit of the published target-band envelope only. It does not convert the sheet into a raw STFT grid and it does not create an STFT recoverability claim over a missing time-frequency amplitude map.

11 Computed From Time-Trace STFT Diagnostic

The sidecar also computes a derived diagnostic from the raw Fig. 2a and Fig. 2b transient-reflectance traces. This diagnostic applies a 10 ps window and 5 ps step, extracts intensity near 3.13 THz with a ± 0.05 THz target band, and compares the derived envelopes to the published Fig. 2d target-band traces.

This diagnostic is explicitly labeled:

`computed_from_time_trace_stft_diagnostic`

It is not raw published STFT data and it is not used to erase the NO_STFT_DATA_FOUND boundary.

Field	Value
Computed diagnostic status	PASS
Source sheets	Fig.2a, Fig.2b
Window / step	10 ps / 5 ps
Target band	3.13 THz +/- 0.05 THz
Traces computed	8
Time points computed	216
Peak records	8
Mean group velocity	101754 m/s
Mean envelope correlation vs. published	0.995318
Mean absolute peak-time delta vs. published	5.84163 ps
Raw STFT grid status	NO_STFT_DATA_FOUND

Thickness	Slope	Group velocity	R^2	Monotonic delay
224 nm	7.5 ps/um	133333 m/s	1.0	1.0
96 nm	14.25 ps/um	70175.4 m/s	0.998157	1.0

The raw Fig. 2a/Fig. 2b headers do not include thickness labels. The sheet assignment follows the published Fig. 2d propagation peak-time ordering and is therefore recorded as an assignment rule, not as independent metadata discovery.

12 Unified Harmonic-Address Persistence Summary

The final sidecar layer gathers the spectral, published-trace, computed-STFT, and raw-grid-boundary outputs into one component ledger. This is a summary of already bounded outputs, not a new materials theory.

Field	Value
Persistence summary status	PASS
Mode	ferron_harmonic_address_persistence_summary
Target frequency	3.13 THz
Combined persistence proxy	0.957192
Raw STFT grid status	NO_STFT_DATA_FOUND
Raw grid boundary preserved	True

Component	Value	Included	Source
target_peak_presence	1.0	yes	spectral audit
spectral_recoverability	0.928455	yes	spectral audit
no_sustained_k_star	1.0	yes	spectral audit
published_propagation_monotonicity	1.0	yes	published trace audit
published_velocity_consistency	0.794118	yes	published trace audit
computed_envelope_match	0.995318	yes	computed STFT diagnostic
computed_velocity_consistency	0.982456	yes	computed STFT diagnostic
raw_stft_grid_boundary	None	no	raw STFT audit

The raw-grid boundary is excluded from the numeric proxy. It is a data-availability boundary, not a persistence penalty. The bounded interpretation is:

The ferron sidecar shows a stable target-window spectral feature near 3.13 THz, published target-band STFT intensity traces with monotonic propagation delay, and computed-from-time-trace STFT envelopes that closely match the published traces. The raw STFT time-frequency grid remains unavailable in the downloaded XLSX files; that boundary is preserved. This is a bounded external-data persistence summary, not a proof or materials-theory claim.

13 Figures

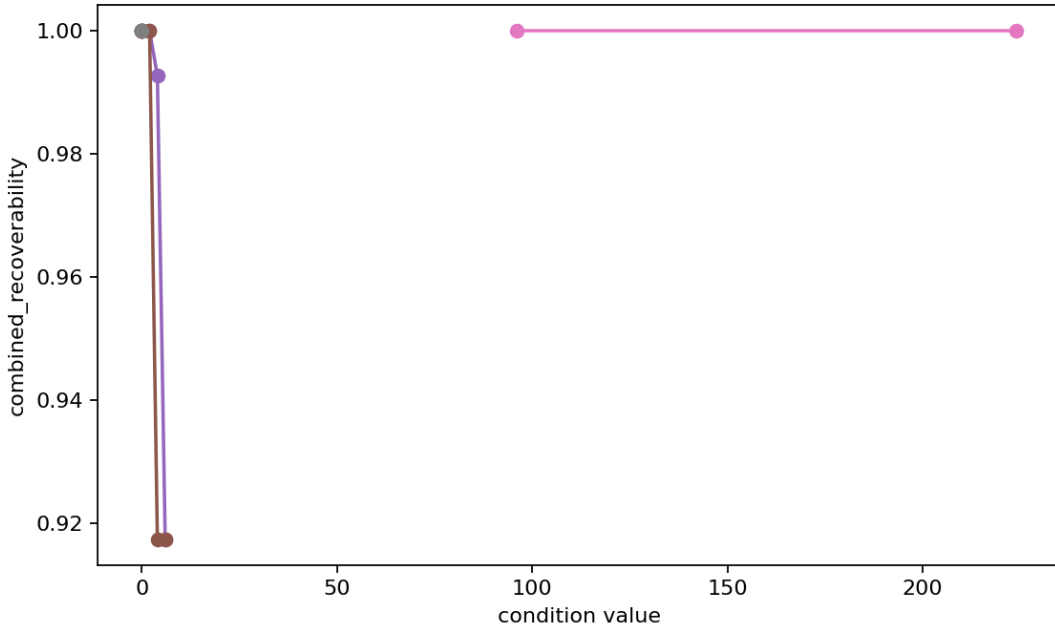


Figure 1: Combined recoverability versus parsed condition. The distance-series traces remain above the configured collapse threshold under the current proxy setup.

14 Interpretation

The correct interpretation is narrow:

HAOS-IIP Materials Bridge Line A is now running on real ferron / NbOI₂ data. The pipeline downloads published Figshare XLSX source data, records provenance, parses time traces and spectra, runs bounded recoverability metrics, and returns **PASS** as an external-data bridge. The second-pass spectral audit finds the target-window peak in 15/15 records with mean peak frequency 3.13893 THz and mean spectral recoverability 0.928455. Under the current proxy thresholds, no k_* collapse is detected. The raw STFT audit finds no validated time-frequency grid, so no STFT recoverability claim is made.

The additional trace and persistence layers refine this without changing the boundary:

The published Fig. 2d target-band traces provide a bounded propagation readout, and the computed-from-time-trace diagnostic reproduces the target-band envelope closely enough to serve as a derived reproducibility check. The unified persistence proxy 0.957192 is therefore a compact sidecar summary of peak presence, spectral recoverability, monotonic propagation delay, velocity consistency, and computed envelope match. It is not a proof of HAOS-IIP, not a claim that ferrons embody HAOS-IIP, and not a raw STFT-grid claim.

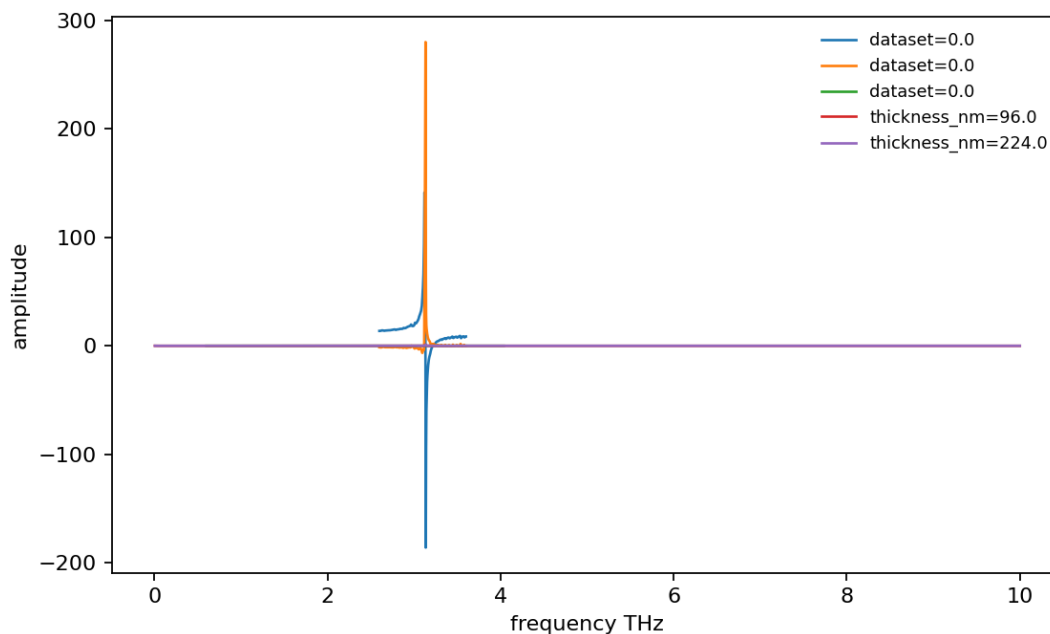


Figure 2: Example parsed spectra from the public Figshare source-data files.

This matters because the result is not tuned into a dramatic claim. External data comes first. Interpretation comes second.

15 Artifacts

Artifact	Path
Loader	experiments/materials_bridge/ferron_coherence_demo/ferron_data_loader.py
Metric model	experiments/materials_bridge/ferron_coherence_demo/ferron_recoverability_model.py
Spectral audit	experiments/materials_bridge/ferron_coherence_demo/spectral_feature_audit.py
STFT audit	experiments/materials_bridge/ferron_coherence_demo/stft_feature_audit.py
Computed STFT diagnostic	experiments/materials_bridge/ferron_coherence_demo/ferron_computed_stft_diagnostic.py
Persistence summary	experiments/materials_bridge/ferron_coherence_demo/ferron_harmonic_address_persistence_summary.py
Runner	experiments/materials_bridge/ferron_coherence_demo/run_ferron_coherence_demo.py

Artifact	Path
Manifest	experiments/materials_bridge/ferron_coherence_demo/data_manifest.json
Validation report	experiments/materials_bridge/ferron_coherence_demo/ferron_coherence_validation.md
Results CSV	experiments/materials_bridge/ferron_coherence_demo/outputs/results.csv
Spectral summary	experiments/materials_bridge/ferron_coherence_demo/outputs/spectral_feature_summary.json
STFT summary	experiments/materials_bridge/ferron_coherence_demo/outputs/stft_feature_summary.json
Published target-band trace summary	experiments/materials_bridge/ferron_coherence_demo/outputs/published_stft_trace_summary.json
Computed STFT summary	experiments/materials_bridge/ferron_coherence_demo/outputs/computed_stft_diagnostic/computed_stft_summary.json
Persistence summary JSON	experiments/materials_bridge/ferron_coherence_demo/outputs/ferron_harmonic_address_persistence_summary.json
Summary JSON	experiments/materials_bridge/ferron_coherence_demo/outputs/summary.json

16 Conclusion

Release 64.1 adds a real external materials-data bridge to HAOS-IIP. It is a bounded operational milestone, not a theory escalation. The ferron / NbOI₂ source data loads, parses, hashes, plots, and produces recoverability telemetry. The measured spectral result is stable under the current proxy thresholds: no k_* , no visible failure, and no early-detection claim. The published target-band STFT intensity traces add a bounded propagation readout, and the computed-from-time-trace diagnostic provides a derived reproducibility check against those traces. The unified persistence proxy is 0.957192. The raw STFT audit also says no cleanly: no validated raw STFT/time-frequency grid is present in the downloaded XLSX files, so no raw STFT-grid recoverability claim is made. That non-hyped outcome is the point of the bridge.

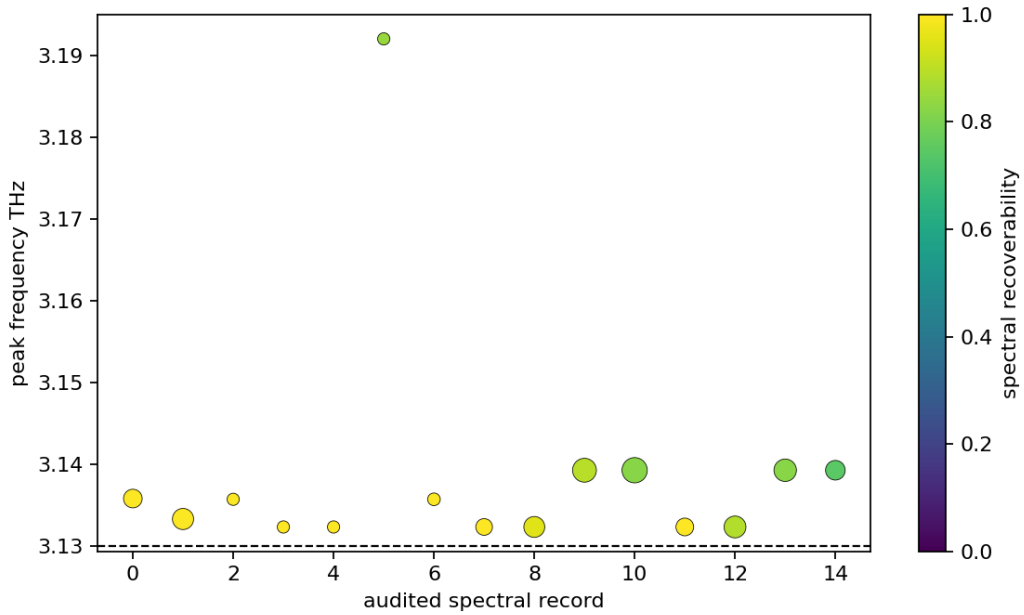


Figure 3: Second-pass spectral peak audit. The target-window peak remains near 3.13 THz across the audited records.

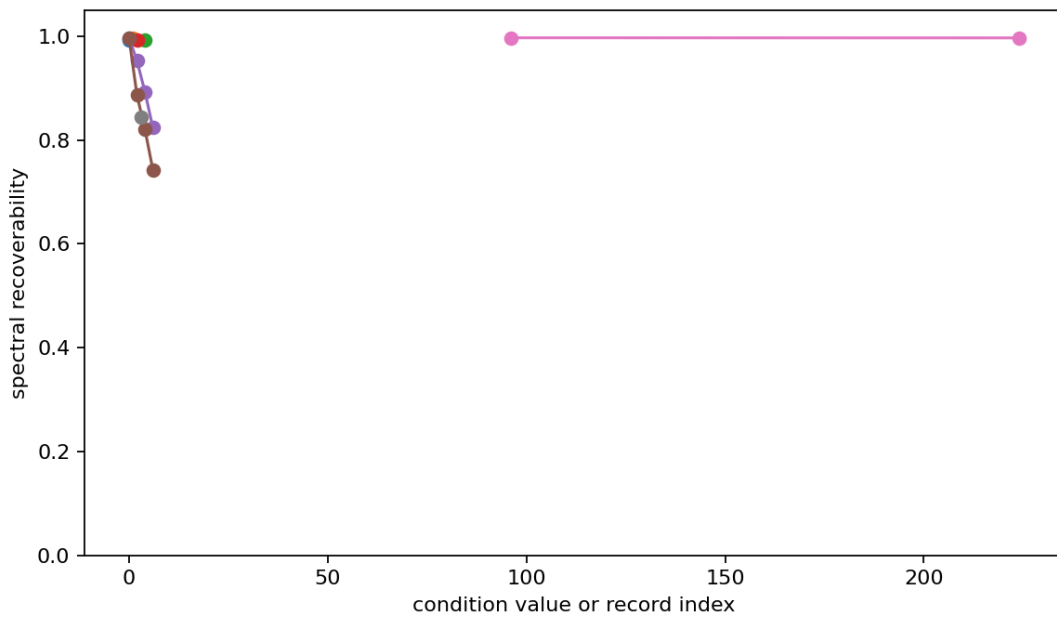


Figure 4: Spectral recoverability audit across parsed conditions. No sustained $k_\star$ collapse is detected under the current proxy thresholds.

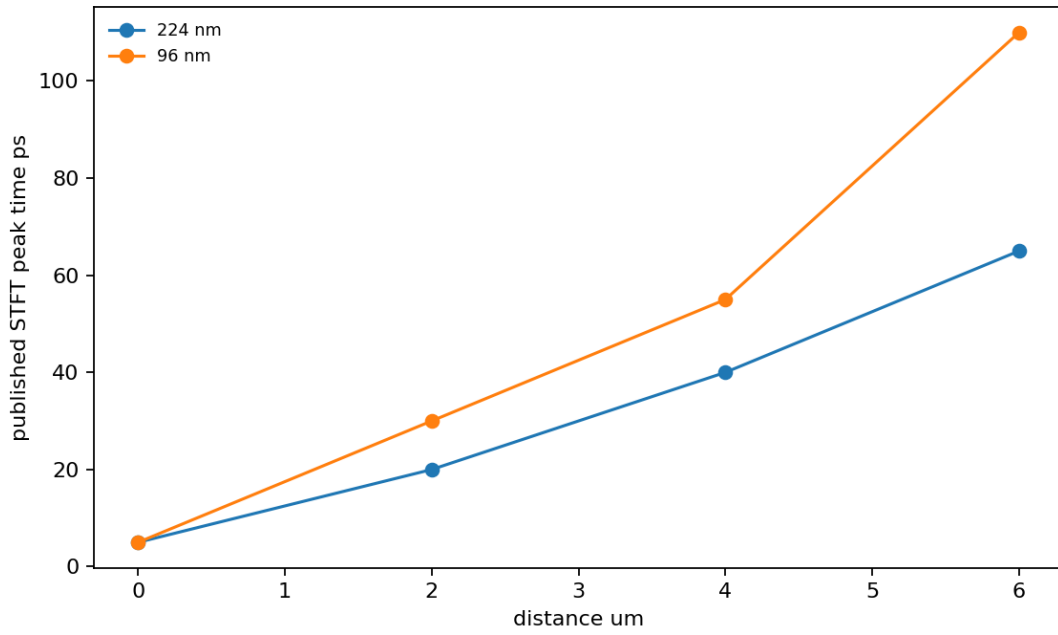


Figure 5: Published Fig. 2d target-band STFT intensity traces: peak time versus distance. These are post-processed target-band traces, not raw time-frequency grids.

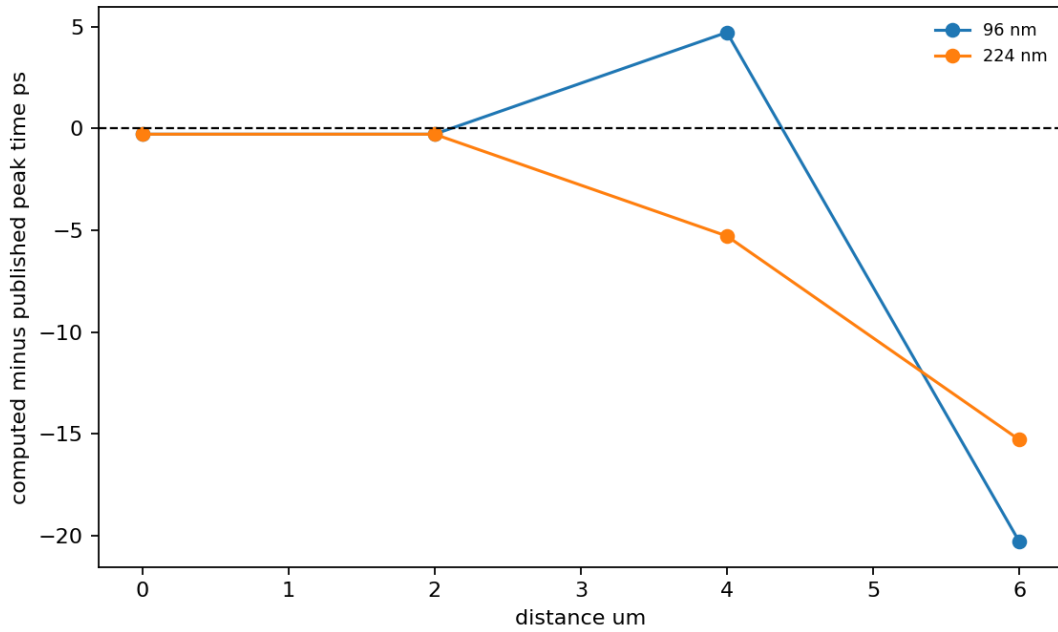


Figure 6: Derived computed-from-time-trace STFT target-band envelopes compared with the published Fig. 2d target-band traces.

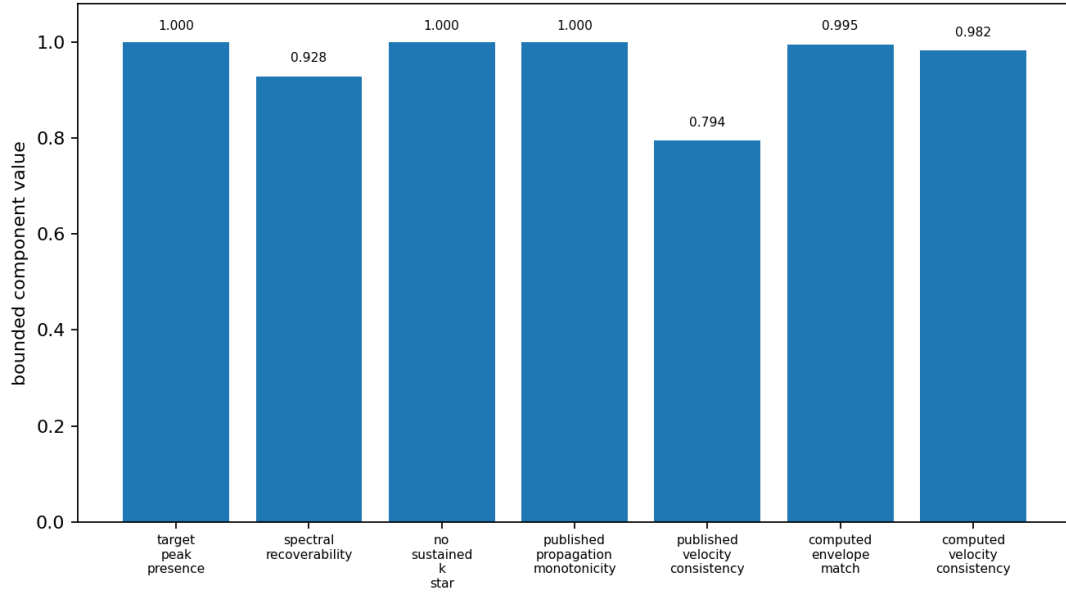


Figure 7: Unified persistence summary components. The raw STFT grid boundary is recorded separately and is not included as a numeric persistence penalty.